



Reservoir computing and non-linear dynamics for time series analysis: An application in the financial market

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ABSTRACT

In various time series analysis scenarios, especially when some type of forecasting is intended, a pre-analysis of volatility, seasonality, and other data characteristics is recommended before the use of a forecasting model. This is a common scenario, for example, in the financial market. In this sense, this research aims to develop a mathematical-computational solution at two levels. In the first one, non-linear dynamics techniques are applied. These are incorporated here through the Hurst exponent, so that the series are grouped and combined with this measure. The purpose here is to extract different characteristic patterns present in this non-linear dynamics metric. Next, a reservoir computing (RC) model is applied to each combination independently, aiming to obtain a more robust general system capable of significantly improving its performance compared to the original RC model and other state-of-the-art predictive techniques. We expect that the proposed model will be able to extract information on long-term dependence, trends, as well as persistence and antipersistence patterns present in the data, which are incorporated through the Hurst exponents. Such additional information is employed here to improve the forecasting capacity of the model.

1. Introduction

In recent years, interest in time series forecasting has grown significantly in various fields, such as medicine [1], meteorology [2], physics [3], epidemic propagation [4–6] and many others. The ability to perform accurate forecasting using machine learning [2,7] and deep learning methods [6,8] has been extensively explored in the literature, demonstrating its potential in various domains.

In the financial context, time series forecasting has become particularly relevant, especially for investment decision-making. Accurately predicting movements in financial values is of great importance to investors and has become a growing focus of study. As we advance in the exploration of machine learning and deep learning algorithms, it is clear that there is significant potential for financial improvement, especially when compared to traditional methods [9]. Furthermore, the combination of deep learning with classical approaches has shown promise, especially when dealing with more complex forecasting tasks.

Understanding aspects of time series, such as their nonlinear dynamics, becomes a crucial tool in this field. In this context, specific tools used to study the evolution of complex systems have been improved, adapted and play an important role. Nonlinear dynamics techniques,

in particular, provide a useful means for analyzing time series from this perspective. This well-established area of science is well suited to capture the complexity of time series, providing a broader view of this type of data.

In this scenario, we explore the application of nonlinear dynamics concepts in time series forecasting, with a special focus on the Hurst Exponent [10], which reveals the presence of (anti)persistent trends in the data. Furthermore, the proposed method combines this analysis with Reservoir Computing, a recurrent machine learning technique that has recently gained prominence in the analysis of time series associated with dynamical systems. In this sense, the proposed methodology aims to implicitly interpret the time series as a complex dynamic system both in its representation (through the Hurst exponent) and in the predictive model. Therefore, this is a natural choice to serve as a predictive algorithm in this type of modeling. In terms of application, the methodology developed here focuses on the financial market. Given the relative ease of access to large amounts of data, we focus on the problem of forecasting financial investments around the world, such as the S&P500 (Standard & Poor's 500 Index), which is the financial market index composed of the 500 largest companies in the United

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States, Dow Jones Industrial Average (DJIA), Hang Seng Index (HSI), New York Stock Exchange (NYSE), Nasdaq Composite Index (NASDAQ), Financial Times Stock Exchange (FTSE) among others.

The major contributions presented in this study are the following:

1. We introduce a novel architecture based on the reservoir computing paradigm, using Hurst exponents to identify and quantify the chaotic nature of time series. This results in a richer and more informative representation of the data, improving in this way the forecasting ability, especially for time series that exhibit nonlinear dynamics behavior, which is usual for example in financial market.
2. We present a flexible pipeline, which despite the focus here on finance, can be easily adapted to a wide range of time series forecasting scenarios, such as demand forecasting, traffic forecasting, or other dynamic processes.
3. We conduct experimental validation on a range of real-world datasets, consistently outperforming baseline and state-of-the-art models in terms of error metrics (MAE, MSE, RMSE, MAPE, and R^2).

The remainder of this paper is organized as follows. Section 2 presents the related works. Section 3 presents the theoretical foundation, within which are the subsections on Hurst Exponents and Reservoir Computing. Section 4 presents the proposed methodology. Section 5 presents the experimental setup, within which are the subsections on evaluation metrics and datasets. Section 6 presents the results and discussions. And Section 7 presents the conclusions.

2. Related works

For the task of time series forecasting, in general, we can mention traditional models, for example, exponential smoothing [11], designed to handle univariate time series, and the Autoregressive Integrated Moving Average (ARIMA) method [12]. ARIMA incorporates an integration idea to deal with non-stationary time series. Among traditional approaches, we can also mention autoregressive (AR) models [13], moving average (MA) [9], and the popular GARCH (Generalized Autoregressive Conditional Heteroskedasticity) method [14].

Although these techniques provide acceptable results in many practical situations, they often face limitations in capturing the non-linear and complex behaviors present in financial markets. In addition to traditional methods, some modern approaches [15,16] have been applied over the years to predict asset prices, volatility, and other financial indicators. For example, the study in [17] introduced a robust deep learning framework using Long Short-Term Memory (LSTM) for stock trading, demonstrating their superiority over traditional strategies. In [18], the authors propose Wavelet Entropy Energy Measure (WEEM), based on the wavelet transform and information entropy, to quantify the intrinsic predictability of time series. The authors in [19] propose a model using a convolutional neural network (CNN), LSTM, and autoregressive moving average (ARMA) model. They combine these models in an ensemble framework to model the mixture of linear and nonlinear data features in the financial time series. In addition, innovative approaches, such as ordinal split networks, have been applied to analyze cryptocurrency price fluctuations, highlighting the potential of complex network methods in the analysis of financial time series [20].

It is also worth noting that in recent years there has been an increase in researchers interest in exploring non-linear dynamics analysis tools to improve time series forecasting. Both chaos theory and non-linear dynamic systems, in general, have provided valuable insights into the complexity of these financial series. Metrics such as fractal dimension [21], Lyapunov exponents [22], and Hurst exponents [10] are some of the techniques that stand out in this type of approach.

Echo State Networks (ESNs), a specific class of recurrent neural networks belonging to the family of algorithms known as Reservoir

Computing, have demonstrated potential in time series forecasting. These networks are particularly efficient in capturing long-term information and complex relationships in temporal data. Studies such as those by Jaeger and Haas [23], pioneers in introducing the ESN framework, have demonstrated their ability to solve nonlinear problems. More recently, the study by Jordanou et al. [24], which applied Echo State Networks to practical nonlinear model predictive control of unknown dynamic systems, has reinforced the potential of these networks in time series forecasting across various domains.

In the financial context, the use of ESNs has also advanced, as demonstrated by Tan et al. [25], who applied ESNs to predict stock prices and achieved superior results compared to traditional models. Studies such as Tian and Pei [26] highlight the use of Echo State Networks in stock price prediction, demonstrating their ability to capture non-linearity and handle the high volatility of financial data.

At the same time, the integration of nonlinear analysis methods, such as fractal dimension [27], Hurst, and Lyapunov exponents [28], has been increasingly explored in time series forecasting. Recent works, such as Escot and Sandubete [29], investigate the use of the Lyapunov exponent to understand stability and chaotic behavior in time series, particularly in noisy environments, where they propose global and local Jacobian indirect regression methods for more accurate estimation.

In finance, the application of this exponent was explored in [30], focusing on identifying patterns of chaotic behavior in financial indices, specifically in predicting large declines in the S&P 500 index using the moving Lyapunov exponent. As in [31], which has the application of the Hurst exponent in the financial market, as well as the improvement of efficiency and precision in its estimation.

Nevertheless, most of these studies in the literature limit their analysis to a single aspect of nonlinear dynamics, without exploring the different approaches of dynamic analysis in combination with predictive models also inspired by nonlinear dynamics, as is the case of ESNs. In this context, the proposed work seeks to fill this gap by combining measures such as the Hurst exponent in a hybrid Reservoir Computing approach to improve the accuracy of forecasting financial time series.

3. Theoretical background

3.1. Hurst exponent

Hurst exponent is a measure of persistence or anti-persistence in time series. It is employed to assess the existence and stability of trends in a dataset. The method used here to compute the Hurst exponent is a technique called the *rescaled range* (R/S). This technique involves computing the range of variation in a time series over different time intervals and comparing these ranges with the average size of the data range.

Consider the time series $X = \{X_1, X_2, \dots, X_n\}$. The steps to compute the Hurst exponent of this series using the R/S analysis are described in the following:

1. Compute the mean value X_m from the values $X_1, X_2, \dots, X_n$:

$$X_m = \frac{1}{n} \sum_{i=1}^n X_i \quad (1)$$

2. Compute the adjusted series y by the mean of the time series :

$$Y_t = X_t - X_m \quad t = 1, \dots, n \quad (2)$$

3. Obtain the series Z of cumulative deviations:

$$Z_t = \sum_{i=1}^t Y_i \quad t = 1, \dots, n \quad (3)$$

4. Obtain the range series R :

$$R_t = \max(Z_1, \dots, Z_t) - \min(Z_1, \dots, Z_t) \quad t = 1, \dots, n \quad (4)$$

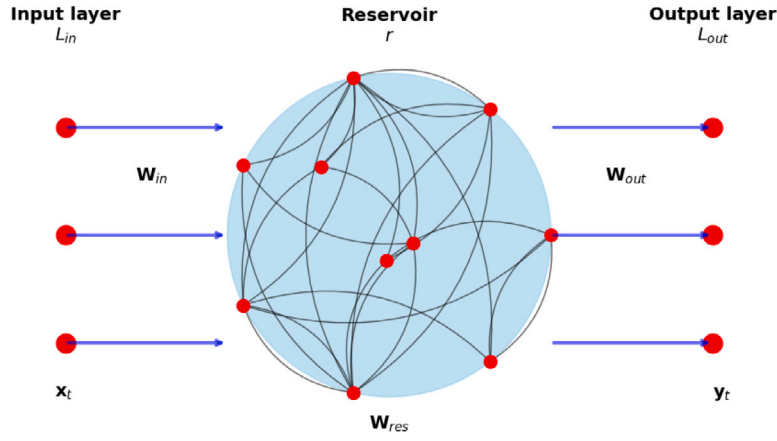


Fig. 1. Structure of the Reservoir Computing model, highlighting its key components: the input layer, dynamic reservoir, and output layer. The input layer receives the preprocessed time series data, which is then fed into the dynamic reservoir, where the model's internal state evolves over time. The output layer maps the reservoir's state to the predicted values.

5. Compute the standard deviation S :

$$S_t = \sqrt{\frac{1}{t} \sum_{i=1}^t (X_i - u)^2} \quad t = 1, \dots, n, \quad (5)$$

where u is the mean of the time series within the interval $X_1 - X_t$.

6. Obtain R/S measure by dividing the magnitude of the cumulative time series by the standard deviation:

$$(R/S)_t = R_t / S_t \quad t = 1, \dots, n. \quad (6)$$

The relationship (R/S) increases according to a power law as time increases, indicating that

$$(R/S)_n = c \times n^H, \quad (7)$$

where c is a constant.

The equation that allows us to obtain the desired value of H through simple linear regression is given by

$$\log(R_n/S_n) = \log(c) + H \times \log(n) \quad (8)$$

From this linear relation, we obtain the plot $\log(n), \log(R_n/S_n)$, which is referred to as the (R/S) plot. The intercept value will be the estimate for $\log(c)$, and the estimate for the value of H , which is the Hurst exponent, will be the slope of the line.

The Hurst coefficient H varies between 0 and 1. A value of H greater than 0.5 indicates that the series has positive long-term memory, meaning it tends to exhibit persistent patterns that continue over time. On the other hand, a value of H less than 0.5 suggests that the series has negative long-term memory, meaning it tends to exhibit anti-persistent behavior that change frequently over time. This can be interpreted as a mean-reverting pattern, where deviations from the mean tend to revert towards the mean in the future.

3.2. Reservoir computing

Reservoir Computing is a machine learning approach focused on forecasting values in time series and sequential processing tasks. It revolves around constructing a reservoir, which corresponds to a recurrent neural network with a special hidden layer. This reservoir consists of a network of internal nodes, where each node is sparsely connected to others. Sparse connectivity is a distinctive feature of Reservoir Computing. The dynamics of the internal nodes in the reservoir are governed by update equations. Each node processes the current input, its current state, and the states of its neighboring nodes, generating a new internal state. The commonly used activation function is the hyperbolic tangent (tanh), although other functions such as ReLU can also be

employed (see Fig. 1). In the Reservoir Computing model, unlike most other deep learning predictive algorithms, the entire neural network is not trained. Training is achieved solely by updating the weights of the output layer connections, making this approach computationally efficient, with fast training and robustness to perturbations. Fig. 8 illustrates the structure of the proposed reservoir applied to time series prediction. The mathematical pathways presented in this section are extracted from [32].

At time t , the network of nonlinear nodes acts as a reservoir r to fuel the structure of an L -dimensional temporal input vector x_t , generating another M -dimensional temporal output vector y_t . Within the reservoir, the internal nodes are interconnected as a randomly directed network, where connections are established by a weight matrix $W_{res} \in \mathbb{R}^{N \times N}$, with $N = |r|$ corresponding to the reservoir size. The input layer L_{in} is fully connected, linking the vector x_t and the internal nodes to the reservoir. The output layer L_{out} is also fully connected, linking the output vector y_t and the internal nodes to the reservoir. L_{in} and L_{out} are related, respectively, by weight matrices $W_{in} \in \mathbb{R}^{N \times L}$ and $W_{out} \in \mathbb{R}^{M \times N}$. The updated state of the internal reservoir nodes, denoted by r_{t+1} , is influenced by the combination of current input x_t and current states r_t , denoted as r_t . The update of this state is determined by a specific mechanism that controls how information from the input and previous states is combined to form the new state r_{t+1} :

$$r_{t+1} = F(W_{in}x_{t+1} + W_{res}r_t) + b_{in}, \quad (9)$$

where F is the activation function, which can be the tanh function or $ReLU$, for example, and b_{in} is the input layer bias. For a specific node n_i , the state is updated in accordance with its current state s_i^t , the states of its input neighbors $\sum_{j \in I_i} s_j^t$, and the possible contributions from the input $\alpha_i^T(W_{in}x_{t+1})$, namely,

$$s_i^{t+1} = s_i^t + \sum_{j \in I_i} s_j^t + \alpha_i^T(W_{in}x_{t+1}), \quad (10)$$

where α_i^T is the transpose of a vector with the i th element equal to 1, otherwise 0. The output $y(t)$ is given by

$$\hat{y}_{t+1} = W_{out}^t r_t + b_{out}, \quad (11)$$

where, in the output layer, b_{out} is the bias, and W_{out}^t is iteratively updated during the training stage, remaining unchanged during testing once training is complete. The training of W_{out}^t minimizes the loss function between the predictions $\hat{y}_{t+1}$, and the actual values y_{t+1} , defined as:

$$\min_{W_{out}^t} \|\hat{y}_{t+1} - y_{t+1}\|^2. \quad (12)$$

It is worth noting that the weight matrices W_{in} , W_{res} , and W_{out} are initialized randomly (see Fig. 2).

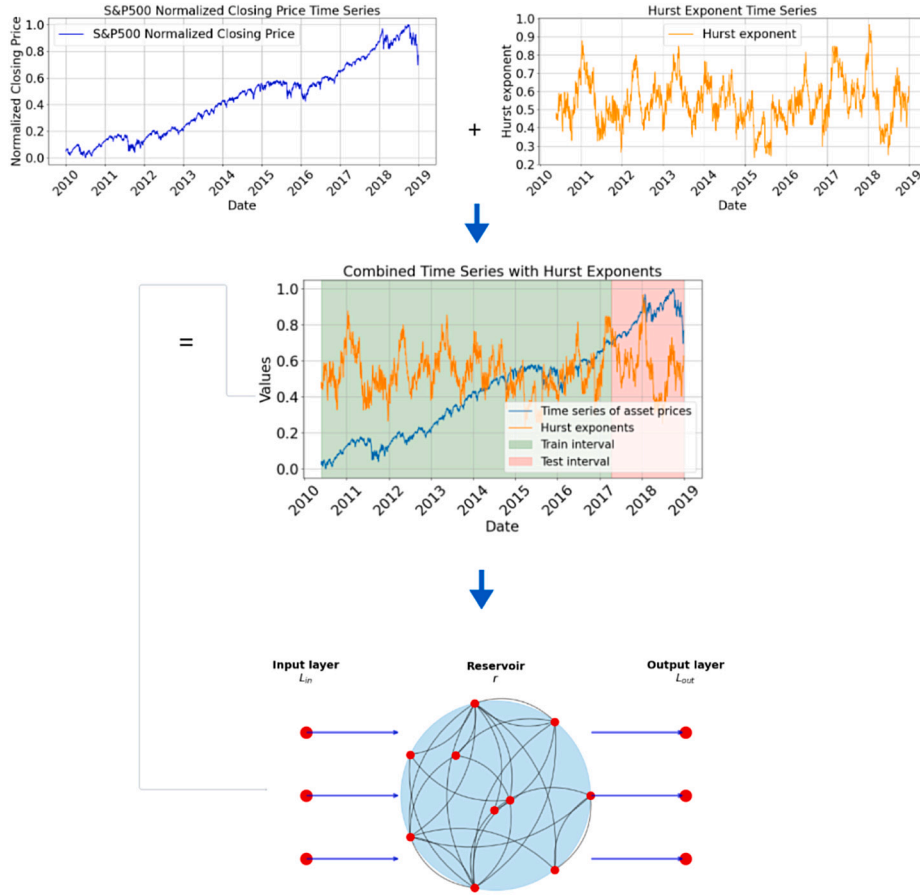


Fig. 2. Structure of the Reservoir Computing model used for time series forecasting. The input consists of concatenated data: normalized time series values and their corresponding Hurst Exponents. By combining these inputs, the model captures short-term patterns and long-term dependencies present in the data.

4. Proposed methodology

In this section, the methodology for predicting the daily closing prices of financial indexes is presented. The source codes are available at the GitHub Repository¹. More specifically, we concatenate the time series data with Hurst exponents computed over those respective data points. Finally, we analyze the model's effectiveness through accuracy metrics, which will be presented in Section 5.1.

For financial forecasting, we incorporate the knowledge of the Hurst Exponents into the RC forecasting model by segmenting the time series of daily closing prices into overlapping 100-day sliding windows. Each window shifts forward by one day, causing each new 100-day segment to overlap the previous one by 99 days. This approach captures the evolving nature of long-term dependency in asset price movements over time. For each of these windows, we compute one value of Hurst exponent, generating at the end a vector of real values denoted H . Therefore, these values are concatenated with the original data and used as input to the RC forecasting model. This process allows the model to learn from important information related to the persistence or anti-persistence of the time series. The concatenation of this nonlinear dynamic measure with the original dataset expands the representation of temporal patterns, thus providing an additional perspective for prediction.

In the RC model training process, the additional information is assimilated by the reservoir's dynamic system, which is a network of neurons with random connections, and this assimilation approach, together with the recurrent nature of the reservoir, allows the model

to capture nonlinear relationships and long-range information present in the input data. After training, we check if the RC model can make accurate predictions for the test data. The data concatenation allows the model to adjust its behavior according to the extra information provided to the forecasting model.

To be more precise, consider a time series denoted by $X = \{x_1, x_2, \dots, x_n\}$, where x_i is the value of the series at the time step i . To create a concatenated representation of the input data for a Reservoir Computing forecasting model, we have the following steps:

1. **Normalization:** First, we normalize the original time series data X to the range $[0,1]$, obtaining the normalized series

$$X_{norm} = \{x_{norm_1}, x_{norm_2}, \dots, x_{norm_n}\}$$

where x_{norm_i} is the normalized value of the series at time i , given by

$$x_{norm_i} = \frac{x_i - \min(X)}{\max(X) - \min(X)}$$

2. **Hurst Exponents:** Next, we compute the Hurst Exponents H_j over the normalized series X_{norm} adopting a 100-day sliding window.
3. **Concatenation:** After computing the Hurst Exponents for each 100-day sliding window, we obtain the series $H = \{H_1, H_2, \dots, H_j\}$, where j corresponds to the total number of 100-day windows fitting entire time series, and H_i ($i = 1, 2, \dots, j$) correspond to the Hurst exponents for each sliding window. We rescale X_{norm} by taking its last j data points to have the same amount of data points as the series H . We concatenate X_{norm} horizontally with

¹ https://github.com/f219979/HURST_RC.git

Table 1

Range of values for the hyperparameters used in the proposed HURST_RC forecasting model. The hyperparameters include the Reservoir size, noise level, Spectral radius, and sparsity, which are critical for tuning the model's dynamics and performance. The selected ranges are based on preliminary experiments aimed at optimizing the model's ability to capture both short-term fluctuations and long-term dependencies in time series data, improving its forecasting accuracy.

Reservoir size	{200, 250, 300, 350, 400, 450, 500}
Sparsity	20%
Noise	{0.01, 0.1}
Spectral radius	{0.7, 0.8, 0.85, 0.9, 0.95}

H to obtain the final input data, denoted by X_{input} :

$$X_{input} = [x_t[i], H[i]],$$

where $x_t[i]$ represents the i th term of the normalized time series, and $H[i]$ represents the corresponding i th Hurst exponent.

4. **RC Forecasting:** The Reservoir Computing forecasting model receives X_{input} as input data and is trained using the algorithm describe in Section 3.2.
5. **Prediction and Denormalization:** After training the model, we use X_{input} to make future predictions of the time series values. At this point, we must denormalize the outputs of the model to obtain the final predicted values.

5. Experimental setup

5.1. Evaluation metrics

To quantify the performance of our predictive model, we adopted several evaluation metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), coefficient of determination (R^2), and Mean Squared Error (MSE). They are formally defined by the following formulas:

$$MAE(\hat{y}_t, y_t) = \frac{1}{T} \sum_{t=1}^T |\hat{y}_t - y_t| \quad (13)$$

$$RMSE(\hat{y}_t, y_t) = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{y}_t - y_t)^2} \quad (14)$$

$$MAPE(\hat{y}_t, y_t) = \frac{1}{T} \sum_{t=1}^T \left| \frac{\hat{y}_t - y_t}{y_t} \right| \cdot 100 \quad (15)$$

$$MSE(\hat{y}_t, y_t) = \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - y_t)^2, \quad (16)$$

$$R^2(\hat{y}_t, y_t) = 1 - \sum_{t=1}^T \left(\frac{\hat{y}_t - y_t}{\hat{y}_t - \langle y \rangle} \right)^2, \quad (17)$$

where $\hat{y}_t$ is the predicted value, y_t is the actual value, and T is the length of the series. For these metrics, the lower the obtained value, the better the predictive performance of the model, with the exception of R^2 , which ranges between 0 and 1, and the closer it is to 1, the better the model predicts the data variation.

5.2. Training/testing split and model configuration

Before applying the prediction model, a preprocessing step is required to ensure the quality and adequacy of the input data. In addition, the configuration of the prediction model must be defined. To achieve this, we split the data into training and testing sets in chronological order, with a ratio of 80% and 20%, respectively. The model performance was evaluated for different combinations of parameters, including: reservoir size, which corresponds to the number of neurons in the hidden layer of the reservoir and influences the model's ability to capture complex dynamics; noise level, which is introduced to improve model generalization by reducing the risk of overfitting;

sparsity, which defines the proportion of zero-weight connections in the reservoir weight matrix, optimizing the representativeness of the connections; and spectral radius, which regulates the largest eigenvalue of the reservoir weight matrix, controlling the stability and dynamic behavior of the model. To improve model stability, the time series data was prepared for analysis through normalization, as shown in Eq. (18).

5.3. Datasets

The data employed in the evaluation of the proposed methodology were collected from Yahoo Finance [33]. In this study, we collected datasets related to the daily closing prices of Standard & Poor's 500 (S&P500), Amazon (AMZN), the Dow Jones Industrial Average (DJIA) index, the Hang Seng (HSI) index, Apple (AAPL), and Tesla (TSLA). The lengths of the collected data vary according to the lengths of the data collected by other existing state-of-the-art research with which we compare our results.

$$X_{norm} = \frac{X_i - X_{min}}{X_{max} - X_{min}}, \quad (18)$$

where X_{norm} corresponds to the time series with normalized data, X_i is the i th term of the time series before normalization, while X_{max} and X_{min} are, respectively, the maximum and minimum values of the time series.

6. Results and discussion

6.1. Results

To demonstrate the effectiveness of the proposed time series forecasting model, we conducted a comparative study with other approaches in the literature. The proposed methodology, where the time series raw data is combined with the Hurst exponents, outperforms the original RC model. For this reason, comparisons with other forecasting models will be made taking into account this combination. To facilitate the nomenclature, we name the RC model using Hurst exponents computed over sliding windows of 100 days as HURST_RC. We also conduct extensive experiments to evaluate the role of different hyperparameters of our model, under different scenarios. Table 1 lists the investigated configurations. For the particular case of sparsity, we tested only one value as the model did not show sensitivity to variations in that hyperparameter.

In [34], we have a forecasting model for S&P500 assets that uses sentiment analysis based on mathematical and deep learning models, more specifically, FinBERT and LSTM transformers. The data was collected from Yahoo Finance from January 30, 2018 to December 30, 2022. We compare this model with HURST_RC using the evaluation metrics MAE, RMSE, MSE, and R^2 .

In [35], performance measures of time series forecasting based on classical machine learning methods are presented, comparing MAE, MSE, and MAPE of ARIMA and LSTM models with the HURST_RC over the period going from March 1, 2000 to June 29, 2020. Note that the price values processed by these methods are not normalized to allow comparison with [35].

In [37], the authors proposed the DAFA-BiLSTM model, which combines elements of vector autoregression (VA) and bidirectional LSTM

Table 2

Comparison of the performance of the proposed HURST_RC model with the SDTP model from [36]. The HURST_RC model significantly outperforms the SDTP model across all indices, achieving lower error metrics and higher R^2 values, showing that the HURST_RC model outperforms the SDTP model for the analyzed indices and time periods.

Models Datasets	HURST_RC				SDTP			
	MAE	RMSE	MAPE	R^2	MAE	RMSE	MAPE	R^2
S&P500	6.33	7.79	0.23	0.9975	14.91	22.78	0.56	0.9808
CAC40	20.01	24.18	0.37	0.9812	31.5	42.25	0.60	0.9478
NYSE	60.44	71.97	0.47	0.9808	63.37	92.50	0.51	0.9713
NASDAQ	35.23	41.55	0.48	0.9952	51.04	75.75	0.72	0.9857
FTSE	16.53	20.77	0.22	0.9923	39.83	52.16	0.54	0.9524
N225	61.25	76.59	0.28	0.9967	159.64	224.86	0.74	0.9751
SSE	13.41	16.88	0.44	0.9964	21.28	29.71	0.70	0.9887
SZ	27.03	34.17	0.24	0.9997	131.65	178.25	1.16	0.9947
SH	8.10	10.21	0.27	0.9979	27.73	31.60	0.73	0.9805
HSI	58.14	74.98	0.22	0.9994	256.02	345.41	0.99	0.9788
DJI	98.68	120.40	0.40	0.9951	145.78	225.80	0.60	0.9842

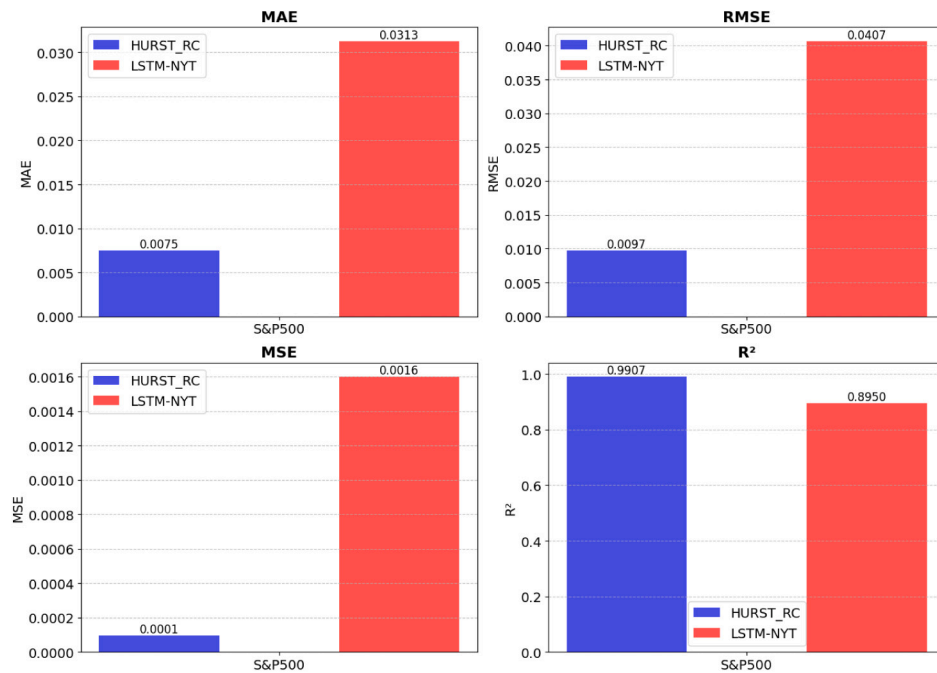


Fig. 3. Bar charts containing evaluation metrics to compare the HURST_RC model with the LSTM-NYT model [34], for the S&P500 index. In this comparison we use the metrics MSE, RMSE, MAE and R^2 , because they are the metrics used to evaluate the effectiveness of the LSTM-NYT model.

networks to forecast the prices of the S&P500 index. The period used to test the model was from 2010 to 2015, and the metrics used for comparison with our model are MAE and MAPE.

In [38], Qiu, Wang, and Zhou (2020) proposed a forecasting model that predicts the stock prices of the Standard & Poor's 500 (S&P 500), Dow Jones Industrial Average (DJIA), and Hang Seng Index (HSI) using a long short-term memory (LSTM) neural network enhanced with an attention mechanism. The data for S&P 500 and DJIA span the period from January 3, 2000 to July 1, 2019, while the data for HSI span the period from January 2, 2002 to July 1, 2019, and the evaluation metrics are MAE, RMSE, MSE, and R^2 .

The comparative results of the proposed HURST_RC model with the models in [34,35,35,37], and [38] are presented in the bar charts in Figs. 3, 4, 5, and 6. We observe that our approach outperformed the literature methods with respect to different metrics in all the evaluated scenarios.

In [36], whose data are presented in Table 2 and in the bar graphs in Fig. 7, the authors describe a model that predicts the asset prices of the S&P500, CAC40, New York Stock Exchange (NYSE), Nasdaq Composite Index (NASDAQ), Financial Times Stock Exchange (FTSE), Nikkei Stock Average (N225), Shanghai Stock Exchange (SSE), Shanghai Composite Index (SH), Shenzhen Component Index (SZ), Hang

Seng Index (HSI), and Dow Jones Industrial Average (DJI), using the Transformer with Period-correlation (SDTP) model. This method employs period-correlation mechanisms and series decomposition layers to better discover relationships between historical series and learn market trends, providing high forecasting accuracy and generalization. The data collection period went from January 4, 2010 to December 28, 2018, for S&P500, CAC40, FTSE, N225, and SSE, and from December 31, 2009 to December 28, 2018 for the other indices.

The bar graphs in Fig. 7 illustrate the results of the comparisons between HURST_RC and SDTP models using the evaluation metrics presented in Table 2.

Tables 3 and 4 show the parameters used for each comparison of the HURST_RC model with the other models presented here. The parameters presented in these Tables include reservoir size, noise level, and spectral radius.

Figs. 8, 9, and 10 show the time series containing the actual and predicted values of the S&P500, CAC40, NYSE, NASDAQ, FTSE, N225, SSE, SZ, SH, HSI, and DJI assets. Fig. 11 displays the time series containing the prices of the CAC40 assets concatenated with the Hurst exponents computed over each 100-day sliding window, which are used as input for the reservoir computing prediction model.

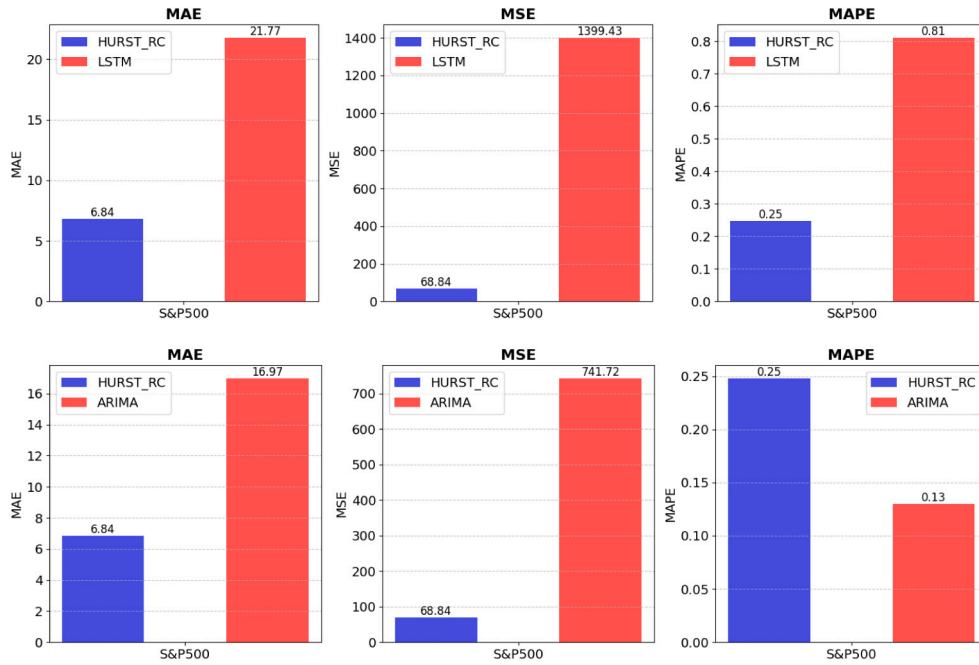


Fig. 4. Bar charts containing evaluation metrics for comparing the HURST_RC model with the LSTM and ARIMA model [35], for the S&P500 index. In this comparison we use the MAE, MSE and MAPE metrics, as they are the metrics used to evaluate the effectiveness of the LSTM and ARIMA models.

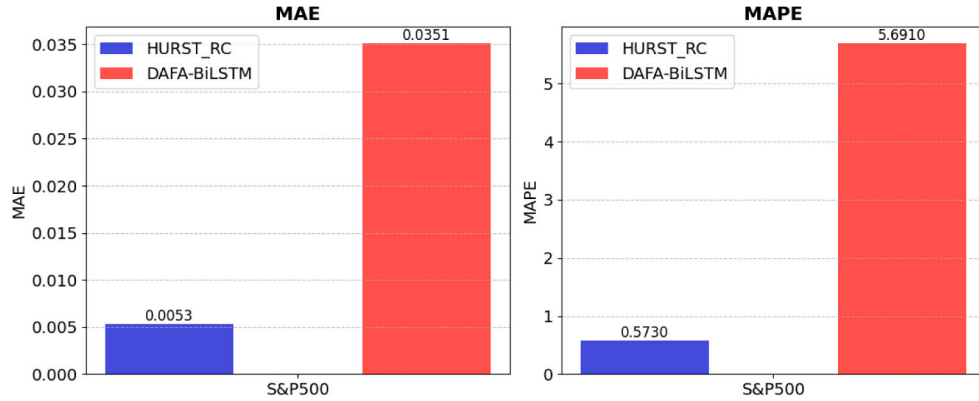


Fig. 5. Bar charts containing evaluation metrics for comparing the HURST_RC model with the DAFA-BiLSTM model [37], for the S&P500 index. In this comparison we use the MAE and MAPE metrics, as they are the metrics used to evaluate the effectiveness of the DAFA-BiLSTM model.

6.2. Discussion

The results presented in the bar graphs of Figs. 3, 4, 5, 6, and 7, and Table 2 confirm the superiority of the HURST_RC model's predictions compared to the other models presented here for predicting financial asset prices. Compared to the LSTM-NYT model, the HURST_RC model obtained gains of 76.03%, 76.16%, 94.06%, and 9.57% in terms of MAE, RMSE, MSE, and R^2 metrics, respectively. Compared to the LSTM model, the HURST_RC model obtained gains of 68.58%, 95.08%, and 69.41% in terms of MAE, MSE, and MAPE metrics, respectively; and when compared to the ARIMA model, the gains were 59.69%, 90.71% and a loss of 47.51% for these same metrics. When compared to the DAFA-BiLSTM model, the HURST_RC model outperformed it by 84.9% and 89.93% in terms of MAE and MAPE metrics, respectively. All of the above predictions were made for the S&P500 index. When compared to the WLSTM + Attention prediction model, the HURST_RC model outperformed it by 94%, 93.58%, 99.59% and 4.14% in S&P500 index predictions; 92.47%, 92.23%, 99.4% and 2.95% in DJIA predictions; 86.87%, 97.05%, 99.91%, and 11.71% on the HSI, for the MAE, RMSE, MSE, and R^2 metrics, respectively. Regarding the

evaluation metrics results in Table 2, which compares the HURST_RC and SDTP forecasting models, the former model outperforms the latter in terms of the MAE, RSME, MAPE, and R^2 metrics, respectively, by 57.54%, 65.8%, 58.92%, and 1.67% on the S&P500 forecasts; 36.47%, 42.76%, 38.33%, and 3.4% on the CAC40; 8.93%, 22.19% and 0.96% in NYSE; 30.97%, 45.14%, 33.33% and 0.95% in NASDAQ index; 58.49%, 60.18%, 59.25% and 4.02% in FTSE forecasts; 61.63%, 65.93%, 62.16% and 2.16% in N225; 36.98%, 43.18%, 37.14% and 0.7% in SSE forecasts; 79.46%, 80.83%, 79.31% and 0.5% in SZ forecasts; 70.78%, 67.68%, 63.01% and 1.74% in SH; 77.29%, 78.29%, 77.77% and 2.06% in HSI forecasts; 32.3%, 46.67%, 33.33% and 1.09% in DJI index.

The presented results suggest that reservoir computing model, when fed with the financial time series data concatenated with the Hurst exponents, outperforms classical and state-of-the-art machine learning and deep learning models in the task of time series forecasting. This suggests that RC model can capture patterns of persistence or antipersistence trends present in the data and assimilate this information effectively, using it favorably to enhance the predictive capacity of the reservoir computing, making it superior to several other predictive models in the literature.

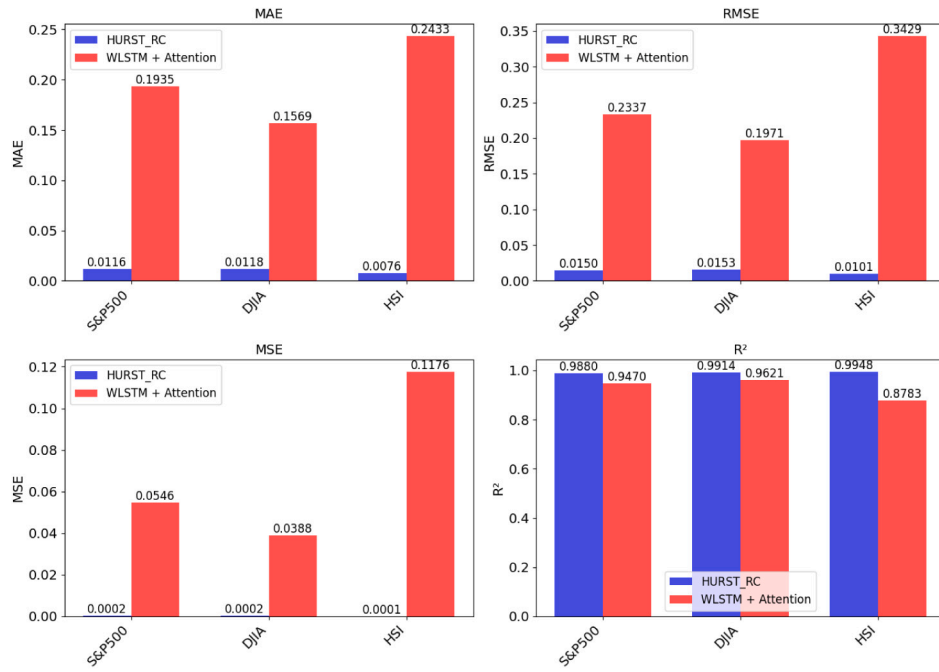


Fig. 6. Bar charts containing evaluation metrics for comparing the HURST_RC model with the WLSTM + attention model [38], for the S&P500, DJIA and HSI indices. In this comparison we use the metrics MAE, RMSE, MSE and R^2 , as they are the metrics used to evaluate the effectiveness of the WLSTM + attention model.

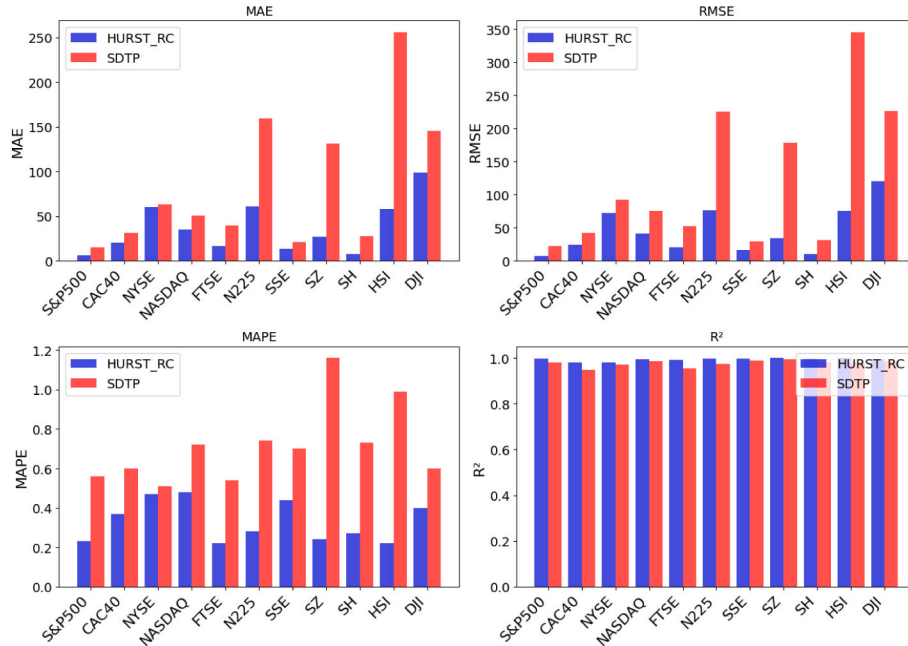


Fig. 7. Bar charts containing the evaluation metrics for comparing the HURST_RC model with the SDTP model for the S&P500, CAC40, NYSE, NASDAQ, FTSE, N225, SSE, SZ, SH, HSI, and DJI indices, respectively. In this comparison we use the metrics MAE, RMSE, MAPE and R^2 , as they are the metrics used to evaluate the effectiveness of the SDTP model.

7. Conclusion

This study demonstrated the effectiveness of the HURST_RC model in forecasting financial asset prices, consistently outperforming a variety of traditional and modern forecasting approaches. For instance, the model achieved significant improvements in evaluation metrics such as MAE, RMSE, MSE, and R^2 . The performance gains were particularly remarkable when compared to the WLSTM + Attention model, where the HURST_RC model showed improvements exceeding 92% in some

metrics for S&P 500 and DJIA forecasts, and over 86% for HSI predictions. Similarly, the HURST_RC model outperformed the SDTP model in indices such as S&P500, CAC40, NYSE, NASDAQ, FTSE, N225, SSE, SZ, SH, HSI, and DJI, as well as other models like DAFA-BiLSTM and LSTM, further emphasizing its robustness and versatility.

These findings highlight the potential of integrating fractal features, such as the Hurst exponent, in time series forecasting. By combining these insights with the power of reservoir computing, the HURST_RC model addresses the challenges posed by the chaotic and nonlinear nature of financial markets, offering an innovative solution.

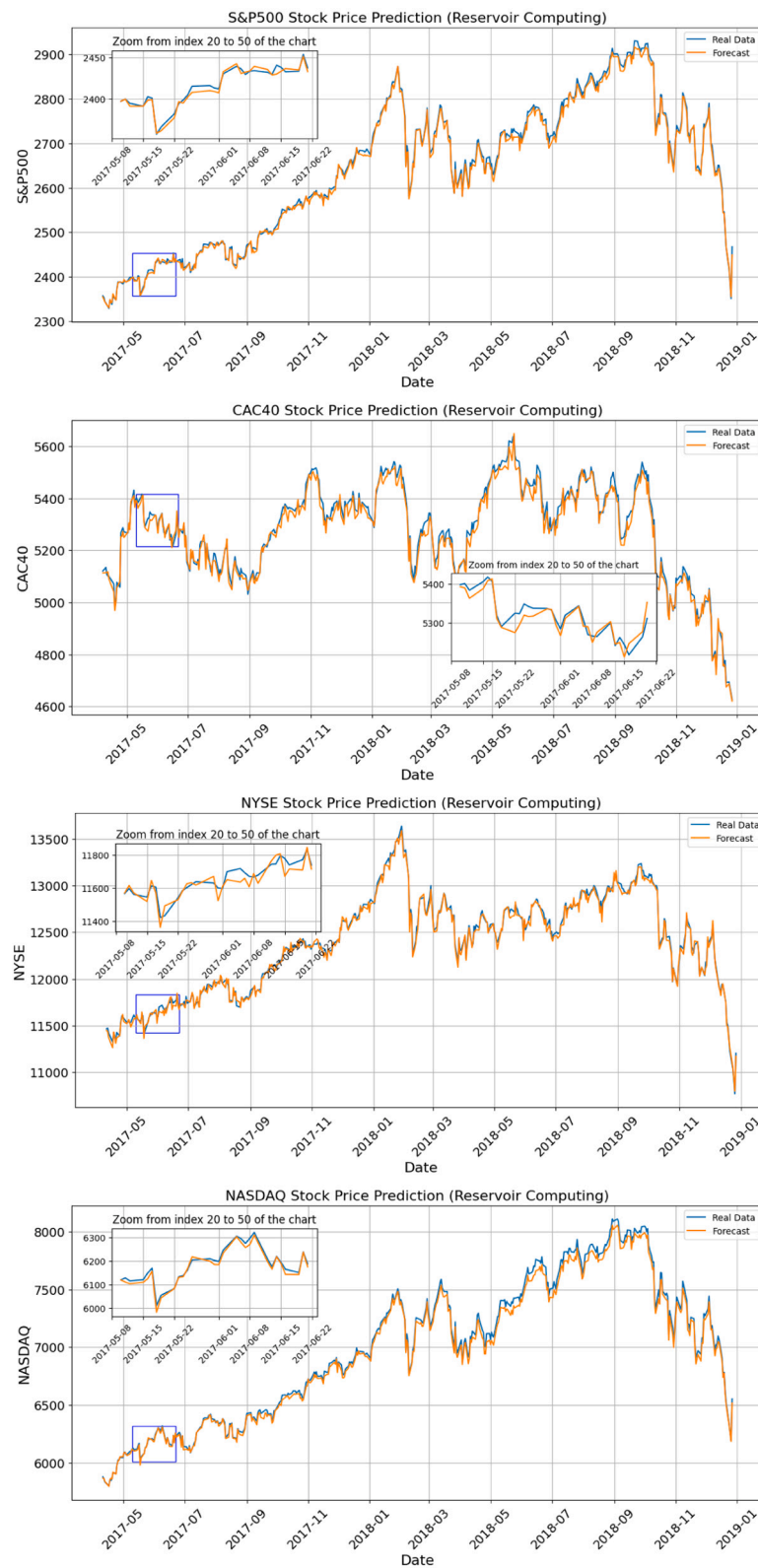


Fig. 8. Time series of actual and predicted values for the indices S&P500, CAC40, NYSE, and NASDAQ.

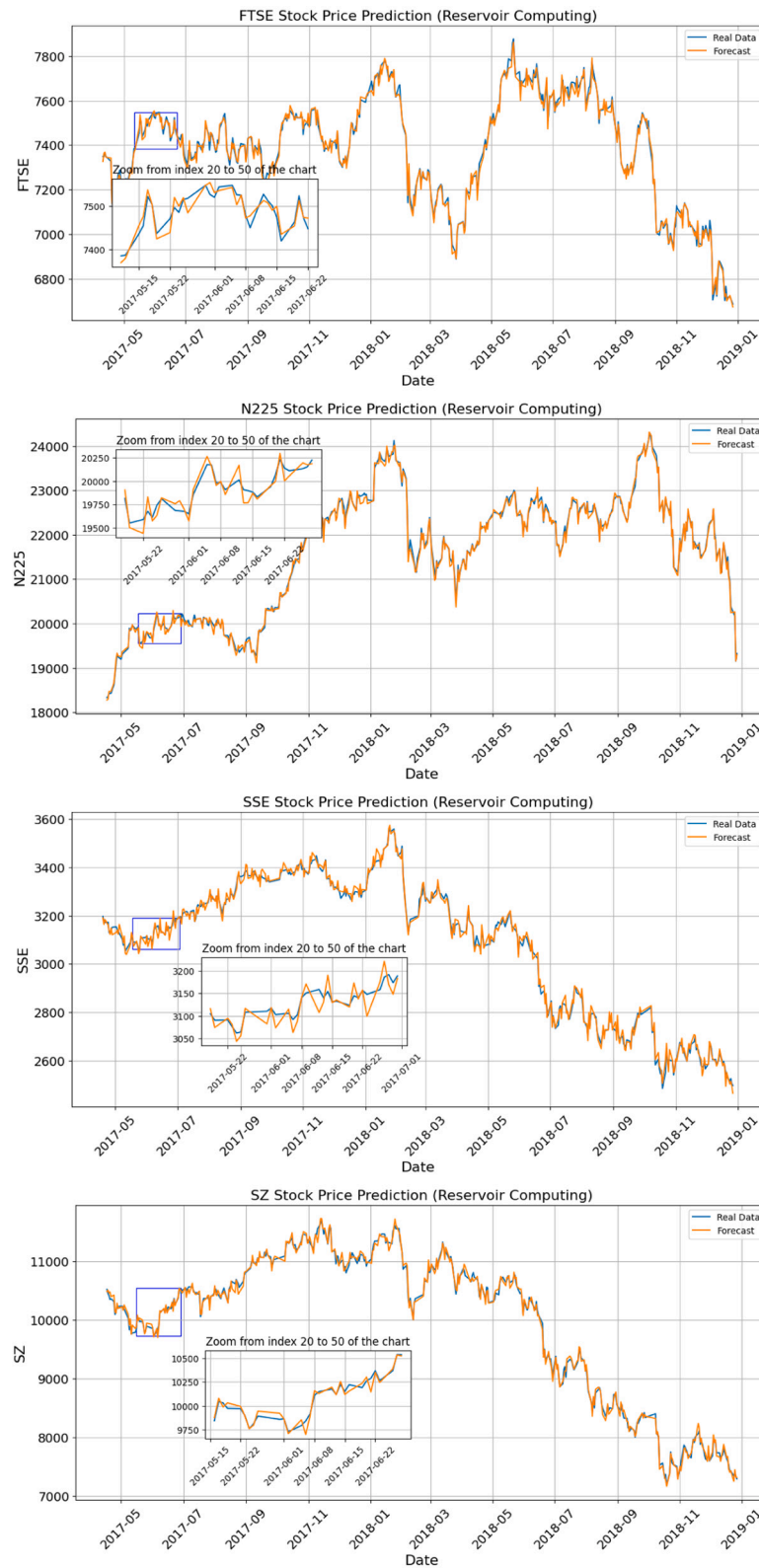


Fig. 9. Time series of actual and predicted values for the indices FTSE, N225, SSE, and SZ.

Table 3

Hyperparameters used in the performance of the HURST_RC model compared with LSTM-NYT, LSTM, ARIMA, DAFA-BiLSTM, and WLSTM + Attention models for financial asset forecasting, whose results are depicted in Figs. 3, 4, 5, and 6.

Experimental settings for comparison with LSTM-NYT	Parameters		
	Noise	Spectral radius	Reservoir size
S&P 500	0.1	0.85	200
Experimental settings for comparison with LSTM and ARIMA	Parameters		
	Noise	Spectral radius	Reservoir size
S&P 500	0.1	0.85	350
Experimental settings for comparison with DAFA-BiLSTM	Parameters		
	Noise	Spectral radius	Reservoir size
S&P 500	0.1	0.9	200
Experimental settings for comparison with WLSTM + Attention	Parameters		
	Noise	Spectral radius	Reservoir size
S&P 500	0.01	0.95	200
DJI	0.01	0.95	200
HSI	0.01	0.95	350

Table 4

Hyperparameters used in the evaluation of the HURST_RC model compared with SDTP for the financial data listed in Table 2.

Experimental settings for comparison with SDTP	Parameters		
	Noise	Spectral radius	Reservoir size
S&P 500	0.1	0.7	200
CAC40	0.1	0.9	200
NYSE	0.1	0.9	200
NASDAQ	0.1	0.85	200
FTSE	0.1	0.85	200
N225	0.1	0.8	250
SSE	0.1	0.9	200
SZ	0.1	0.85	200
SH	0.1	0.85	200
HSI	0.1	0.85	200
DJI	0.1	0.7	200

Looking ahead, this approach could be readily extended to other financial data, such as commodities or cryptocurrencies, or adapted to analyze other time series datasets, beyond the financial area.

CRediT authorship contribution statement

Francisco Alves dos Santos: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Reneé Rodrigues Lima:** Writing – original draft, Visualization, Validation. **Jerson Leite Alves:** Writing – original draft, Visualization, Validation. **Davi Wanderley Misturini:** Writing – original draft, Visualization, Validation. **Joao B. Florindo:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Joao Florindo reports equipment, drugs, or supplies was provided by State of Sao Paulo Research Foundation. Joao Florindo reports financial support was provided by National Council for Scientific and Technological Development. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

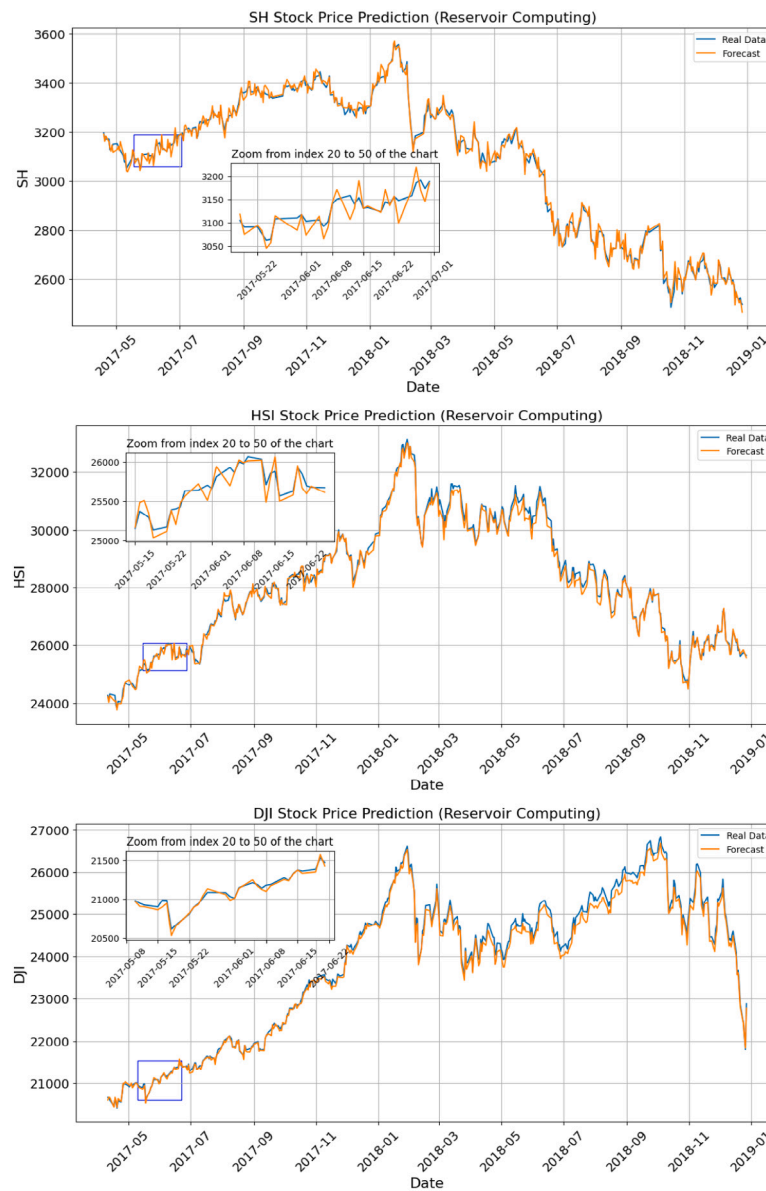


Fig. 10. Time series of actual and predicted values for the indices SH, HSI, and DJI.

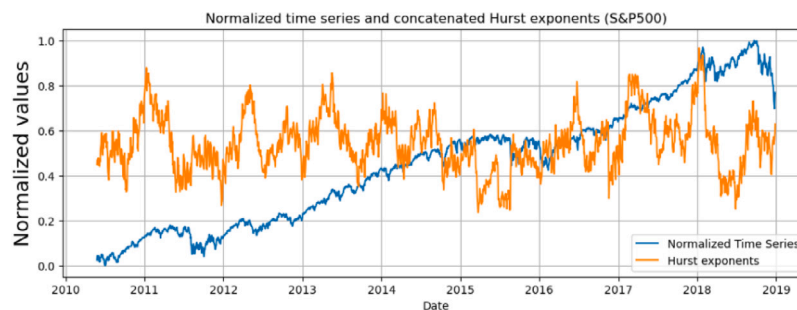


Fig. 11. Time series with the closing prices of S&P500 assets concatenated horizontally with the Hurst exponents computed over each 100-day sliding window, which were used as input for the reservoir computing prediction model.

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